Team ARCANE:
Final Design Review



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Mission Statement

> Team ARCANE aims to create a **supplement to the white cane** that would **warn** visually impaired of **upcoming obstacles** that they might not identify otherwise.

Objectives

A. Conceptual & Planning

A1 Design Camera Mount

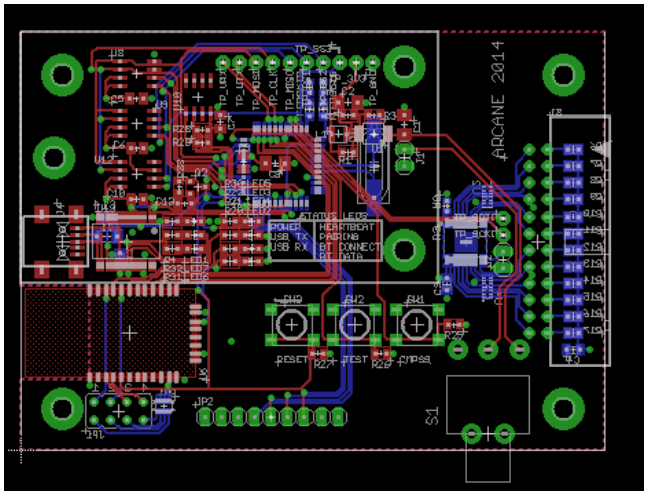
- Designed a **compact enclosure** for the cameras
  - Required second iteration due to time constraint
- Front holes for **lenses**
- Back holes for **wiring**



The camera mount was **successfully created** through a CAD model.

A2 Design Haptic Hardware

- **Integrated PCB** includes Propeller, motor driver, serial over USB and Bluetooth
- Board has headers to **connect compass and battery charger** PCBs



The **design for the PCB** has been successfully created.

Objectives

B. Implementation

B1 Port Depth Map to Intel Board

- OpenCV **optimized for Intel**
  - Uses Intel Integrated Performance Primitives library (e.g. SSE)
- **Intel NUC D54250WYK**
  - Intel i5 4250U
  - 2 cores, 1.3 – 2.6 GHz
  - Wifi + Bluetooth card

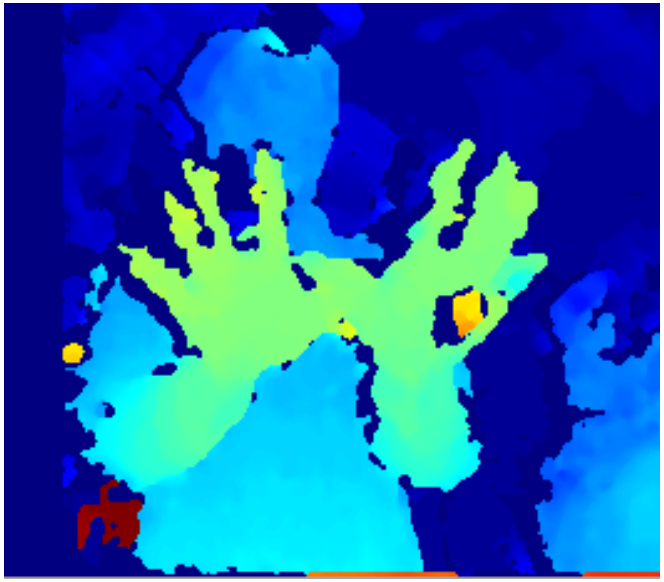


Installed **Ubuntu 12.04LTS** and achieved a **frame rate of over 8 fps.**

B2 Speed Up Block Matching

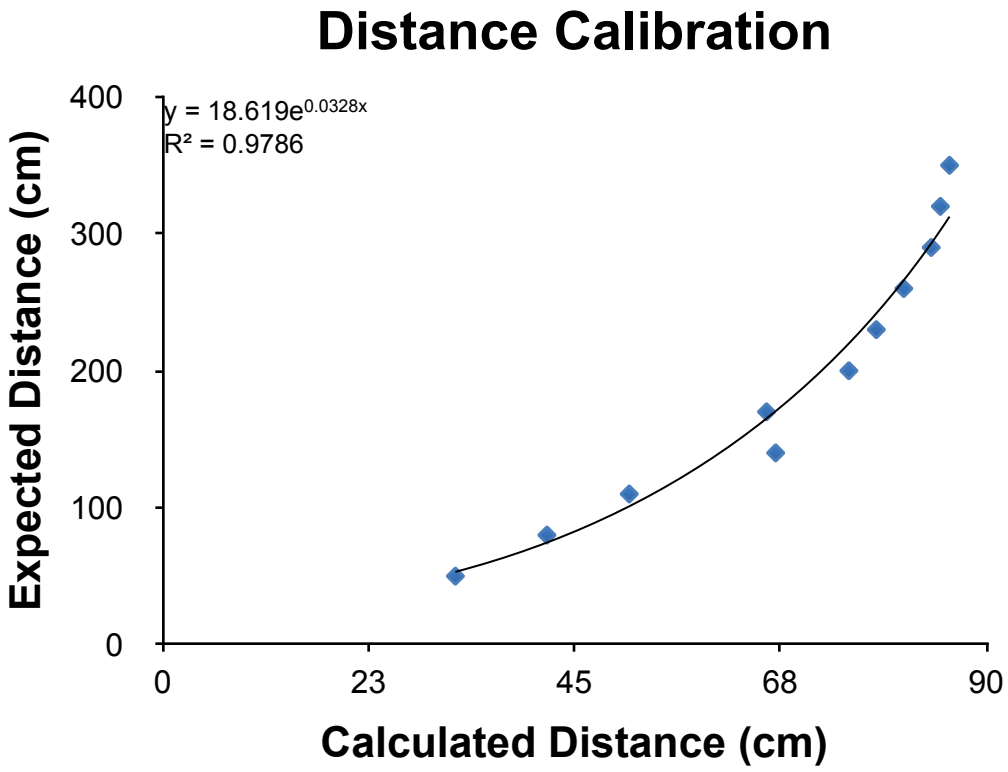
- Changed hardware
- Tweaked **block size**, number of **search paths**, & image **resolution**
- Removed multi-threading

Demonstration



The Semi-Global Block Matching calculation has been **sped up** such that we can generate a **depth map at over 8 fps.**

B3 Accurate Distance Estimation



- **Exponential fit**
- Set **minimum & maximum** values
- Average **percent error of 7.35%**

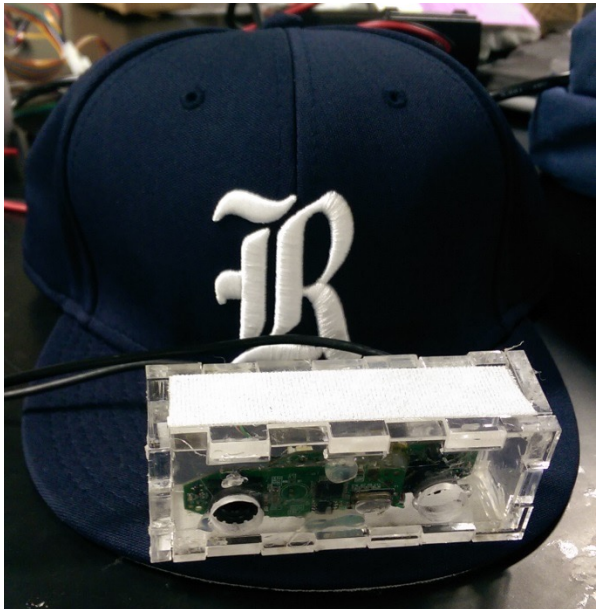
The depth map outputs distance data with an **error margin of 7.35%** of the true distance of the object.

RICE

arcane

B4 Build Camera Mount

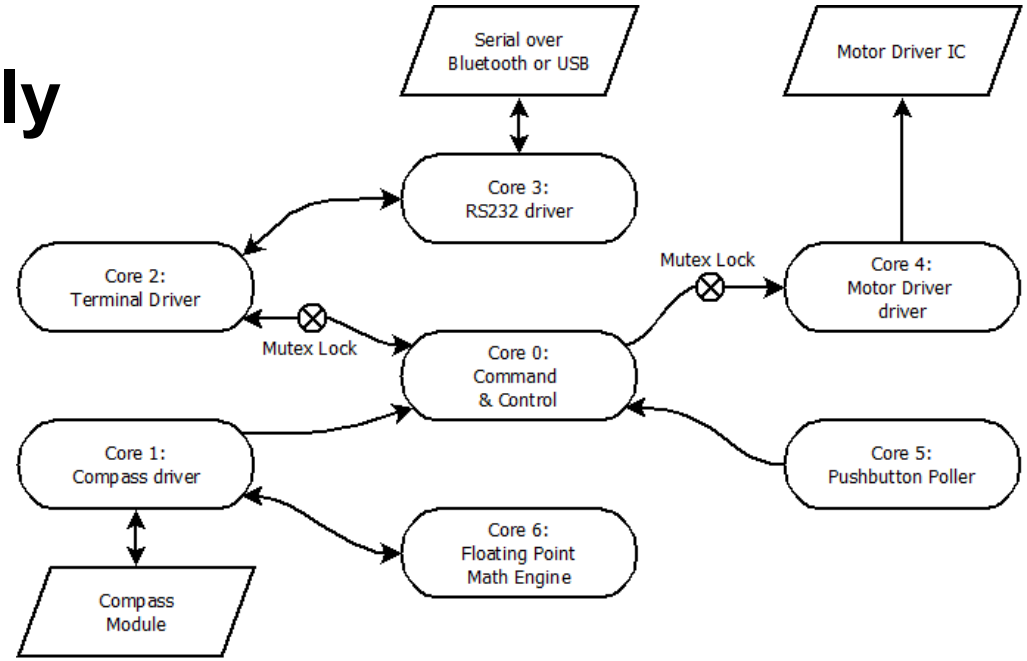
- Enclosure **laser cut and assembled from acrylic**
  - Weather-sealed
- Mount attaches onto a flat bill hat with **Velcro**
  - Does not affect **structure** of hat
  - Brim **does not bounce** as user walks around



A **sealed weather-proof camera mount** has been created that attaches to the **top of any hat.**

B5 Integrate Haptic Software

- Allow compass and sensors to **set motor intensity concurrently**
  - Parallel code
  - Mutex protected
  - 7-way (!) parallel

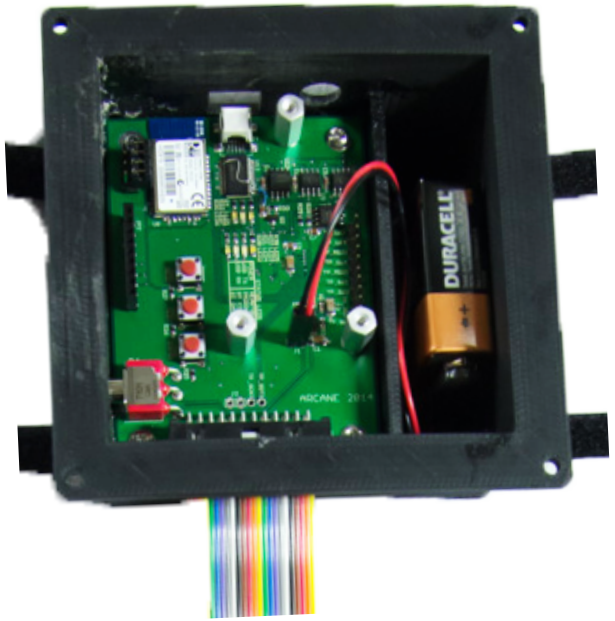


The haptic feedback and compass software **run successfully at the same time.**

B6 Build & Test Subsystem Hardware

- **Successfully built and tested** integrated PCB
- **Bluetooth** module is **prone to ESD**, but nothing else has needed to be replaced

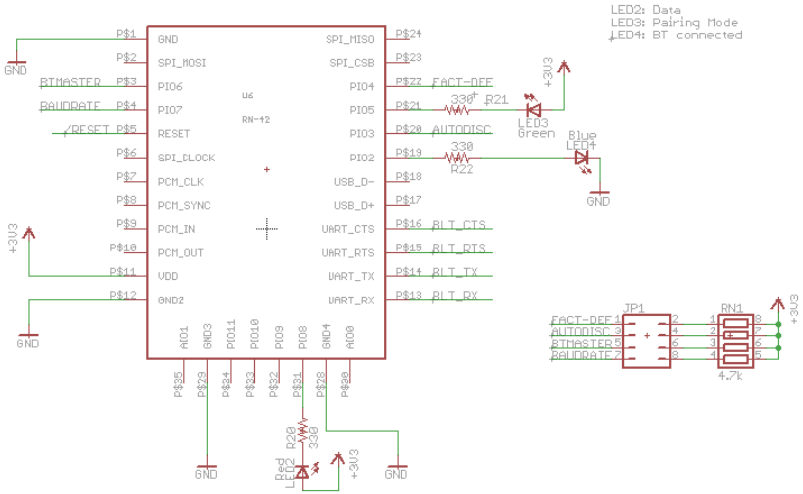
Demonstration



The components on the **integrated PCB work as well** as they did separately.

B7 Bluetooth Communication

- Haptic and sensor subsystems can **communicate over Bluetooth**
- **RN42** on haptic side **interfaces with Propeller** like a serial port

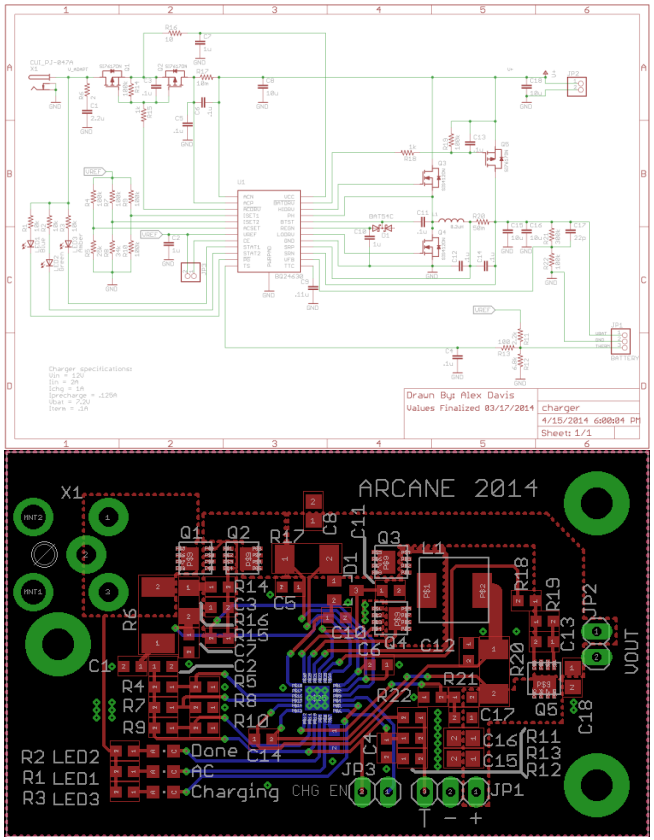


Demonstration

The haptics and sensors subsystems can successfully **communicate through Bluetooth**

B8 Design & Build Power System

- **Haptic system:** LiFePO4 charger and pack
  - Uses TI bq24630 charger IC
  - **~8hr battery life** expected
    - ➢ 7.2Wh battery, .88W avg. load
- **Sensors:** LiFePO4 19.2V 63.4Wh battery
  - **~5hr battery life**



The system is projected to have a **battery life of ~5 hours** with its chargers.

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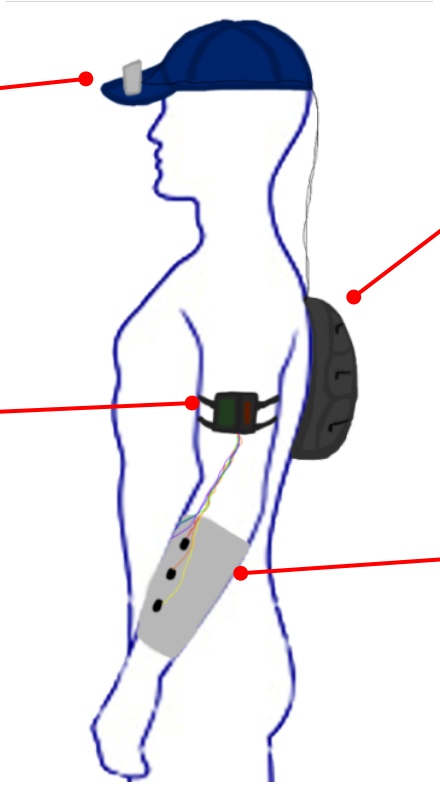
B9 Wearable Final Prototype

One-size-fits-all **cap** for sensor

Backpack for Intel NUC & battery

Bicep attachment for PCB case

Adjustable Velcro arm sleeve for vibration motors



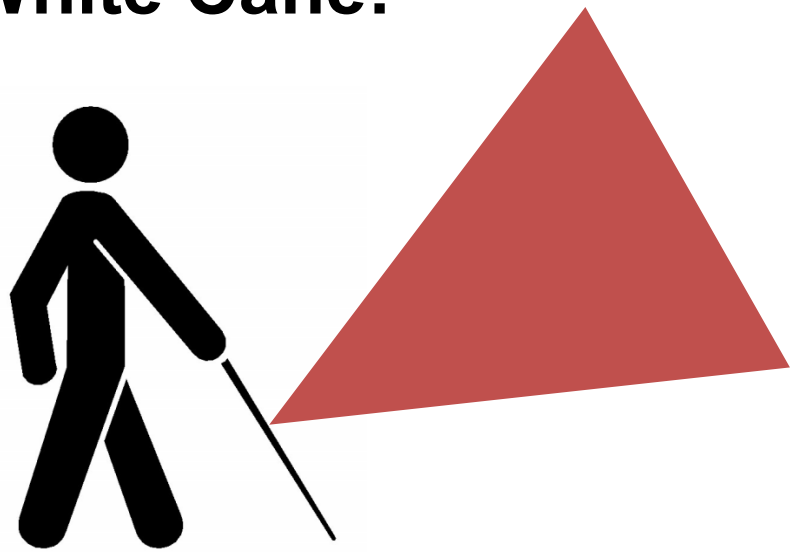
The final prototype **can be worn correctly** by the user with assistance in **~2 minutes**.

Objectives

C. Testing & Verification

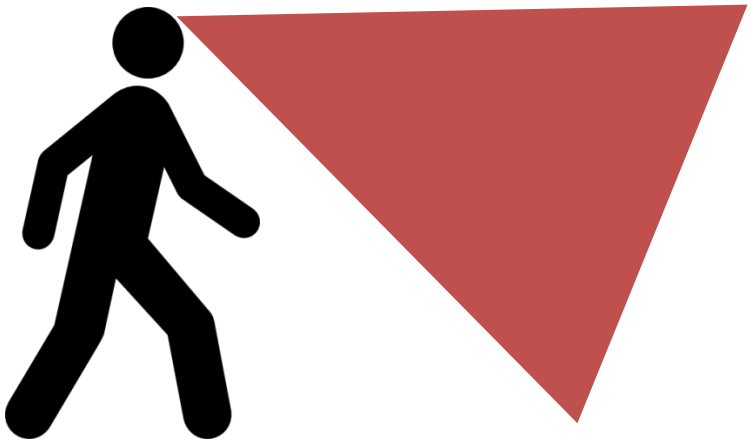
C1 Sensor Placement & Depth Map

White Cane:



- Waved from **side to side**
- Average of **1.28 swing / second**

Head:



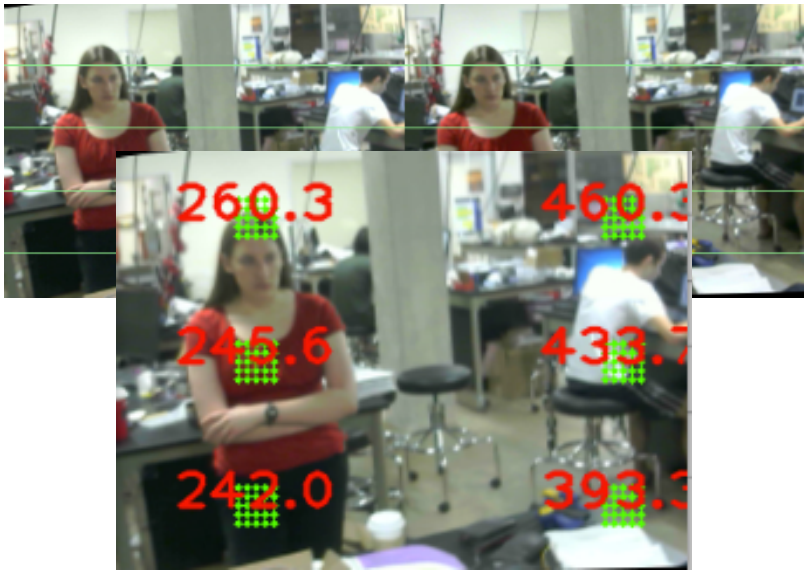
- Head serves as **natural damper**
- Controlled by user

C1 Sensor Placement & Depth Map

White Cane:



Head:



Placing the sensor on the head allows for **more stable image quality,** successfully generating a depth map for the haptic subsystem.

C2 Test on Visually Impaired

- Conducted tests on **seven blind-folded individuals:**
  - 2 minutes of training
  - Vibrated individual motors
  - User identified location
- **78% success rate** of identifying object location
- Users **gained confidence and accuracy** over time

% Accuracy Map of Vibration Layout

93% (26/28)	**83%** (25/30)
86% *(25/29)*	***57%*** *(16/28)*
66% *(19/29)*	***79%*** *(23/29)*

C2 Test on Visually Impaired

- Met with **two visually impaired individuals**
- Feedback on the ARCANE:
  - Hard to **localize vibrations**
  - Suggested **pulses** over constant vibrations
  - Design needs to be **slimmed down**
  - Enjoyed that it **didn't interfere** with daily tasks

While we did have **successful preliminary results** with blind-folded individuals, visually impaired individuals provided us with **constructive feedback** crucial for any future work on this project.

C3 Build Final Prototype

Final integrated system features:

- **Accurate stereovision sensor** on cap
- **Vibratory feedback** to the arm
- **Serial communication** between the two
- **Simple turn on** procedure
- **Easy to put on, wearable** design



Demonstration

Team ARCANE has successfully created a **fully functional final design.**

Acknowledgements

- Fox Family
- Dr. Gary Woods
- Dr. Marcia O'Malley
- Prof. Gene Frantz

OEDK
Oshman Engineering Design Kitchen

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